

Final version accepted in Food Quality and Preference 8/4/22

Food neophobia predicts visual ability in the recognition of prepared food, beyond domain-general factors

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Keywords: Visual ability, object recognition, Food Neophobia, Personality

Author Note

The authors declare no conflict of interest. The construction and validation of the food tasks and food interest survey were part of the second author's Honors thesis in the department of Psychology, Vanderbilt University. The research was supported by the David K. Wilson Chair of Psychology Research Fund at Vanderbilt. We thank Bunmi Olatunji and Conor Smithson for feedback on the manuscript. We provide the materials for both tests free of charge (Gauthier, 2022a, 2022b).

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Abstract

Food neophobia (FN), or the tendency to avoid novel foods, is associated with poor health. Most correlates of FN come from self-report, making it difficult to identify behavioral correlates that may be treatment targets. To begin to fill this gap, we explore the association between FN and the ability to visually recognize images of prepared food. We created the first two tasks specifically designed to measure the visual ability to recognize food and demonstrate their reliability and validity. In a convenience sample of adults, we find that individual differences in FN are negatively related to the ability to recognize images of food, after controlling for a large contribution from domain-general visual recognition ability, and influences of gender and general interest in visual recognition. These results are consistent with the possibility that FN and visual ability to recognize foods could be linked in a feedback loop that is mediated by familiarity. Our work provides tools that should facilitate the exploration of this mechanism, which may maintain the avoidance of novel foods that is commonly observed in FN.

1. Introduction

Many parents worry about their child being a picky eater (Cole et al., 2017) but although avoidance of unfamiliar foods, or food neophobia (FN), can develop during childhood, it can also continue into adult life (Birch, 1999). FN is a stable trait (Januszczyńska & Viaene, 2012) that varies substantially among adults. The Food Neophobia scale (FNS – Pliner & Hobden, 1992), a 10-item instrument based on self-report items rated on 7-pt likert scales, is the gold standard for its measurement. Food neophobia is considered maladaptive behavior with potentially important nutritional and societal consequences (Rabadán & Bernabéu, 2021). FN appears to be partially heritable (e.g., Knaapila et al., 2007) and negatively correlated with education, income and urbanization (e.g., Tuorila et al., 2001). FN can limit healthy eating (e.g., Jaeger et al., 2017) and predicts a diet with lower nutritional quality and a predisposition towards obesity (Sarin et al., 2019). At a societal level, FN could limit the success of efforts to modify food production (Henson, 1995) as a means to improve health or reduce production impacts on climate. For instance, FN limits willingness to eat various types of meat substitutes (De Koning et al., 2020; Siegrist & Hartmann, 2020).

The availability of the FNS has advanced knowledge of the correlates of FN, but much of this knowledge is based on correlation between FNS scores and other self-report scales. These include attitudes driving food choices, basic personality traits such as those measured in the Five Factor Model (Funder, 2001) and other individual characteristics such as gender, education or socio-economic status (Rabadán & Bernabéu, 2021). At the level of individuals, those with high levels of FN are more likely to be influenced by the familiarity of foods in their daily food

choices (Jaeger, Roigard, et al., 2021). However, recent work suggests that novelty of foods may not be the most important factor driving the behaviors measured by the FNS. Arousal induced both by certain novel and familiar foods may be more critical (Jaeger, Chheang, et al., 2021). In the present work, we are not postulating a causal role of visual abilities in FN and our hypothesis that FN may be related to food recognition ability does not depend on a specific model of FN. Indeed, regardless of the factors driving limited experiences with food, this reduced perceptual exposure could result in poorer visual ability in this domain (i.e., poorer scores on visual food recognition tasks). It is possible to conceive of a feedback loop in which FN leads to avoidance of certain foods, and avoidance of some food limits familiarization with novel foods, which in turn is likely to maintain FN. We therefore suggest there may be perceptual correlates of limited experience with food. One result that supports this idea comes from a rare study linking FN to a performance task (as opposed to a self-report measure). In a study of 126 adults (Knaapila et al., 2017), FN was negatively correlated with odor identification scores on a test using spices ($r = -.24$). In the present work, we created visual tests for food recognition ability to test the possibility that FN is likewise related to poorer ability to recognize images of prepared food.

Our efforts in developing the tools to address these ideas is rooted in a program of research on visual abilities, a topic that has experienced substantial growth over the last 30 years. This research recently characterized a domain-general ability, o , defined as a latent variable accounting for the vast majority of the variance across different visual tasks (e.g., learning and matching tasks) and using different categories of novel objects. The ability o generalizes across tasks that use novel or familiar categories (Sunday et al., 2021) and is not

strongly related to general intelligence (Richler et al., 2017, 2019). As such, *g* can predict performance on a task above and beyond factors like experience and general intelligence (Sunday et al., 2018). Critically for this project, an important value of measuring *g* is to control for domain-general abilities, so that we can better isolate a possible domain-specific visual ability for food. In the present study, our first aim was to develop the first tests designed to measure the ability to recognize images of prepared food. Our second goal was then to ask how much of the performance on such tasks was accounted by *g* vs. individual differences specific to the food domain. Finally, we asked whether individual differences in food-specific recognition ability would be related to FN. Demonstrating a link between FN and food recognition ability that is not explained by *g* would provide support for our hypothesis that there may be perceptual correlates of limited experience with food.

Our tests of visual food recognition, like those we have developed in other domains (Chang & Gauthier, 2021; McGugin et al., 2012), do not require the production or matching to verbal labels (unlike the work on odor identification mentioned above) – instead we use matching and oddball decisions that require distinguishing between different dishes that are visually similar. This is because we are specifically interested in a visual ability related to food, which is likely to be correlated with but to some extent dissociable from semantic knowledge in this domain (Barton et al., 2009; Van Gulick et al., 2016). We also include a new survey of attitudes and interest towards food, to ask if any association between FN and food recognition ability may stem from more general attitudes towards food.

Finally, we include a survey of the Five Factor model of personality (McCrae & Costa, 1987). The literature includes evidence of correlations between personality and FN or other

related constructs, such as trait disgust (Al-Shawaf et al., 2015; Knaapila et al., 2011; Nezlek & Forestell, 2019; Tybur et al., 2018). There is very little work measuring personality and individual differences in general visual perception. One study found no relationship between α and any of the five dimensions of personality measured by the IPIP (Richler et al., 2019). It is plausible that personality may have more of a role in specific domains of visual perception, given the links between personality and motivation (e.g., Parks & Guay, 2009). There is some evidence on this question from the domain of face recognition. In this case, extraversion is the dimension most often linked to performance, although the direction of the effect has not been consistent (Kramer, 2021; Kramer et al., 2020; Li et al., 2010). What motivates learning may also differ across specific domains, but this remains an understudied question. As we measure visual ability in the new domain of food and its relation to FN, we wanted to ensure that a link between FN and food recognition ability was not simply supported by more domain-general traits. For instance, FN negatively correlated with Openness in one study (Knaapila et al., 2011). If people high in FN had poorer food recognition ability, we would want to know whether this effect survives controlling for Openness.

Based on work in other domains (Chang & Gauthier, 2021; Chow et al., 2021; Sunday et al., 2018) we expect to find support for a domain-specific ability for the recognition of prepared food. However, food is an interesting domain in which to measure visual recognition ability for a few reasons. First, compared to other domains in which individual differences are expected due to training, such as medical images, musical notation or bird identification, the baseline of experience with food should be particularly high for all participants. Even the least food-oriented adult has seen several thousand instances of prepared food, in person, in menus, in

visual entertainment and advertisements. This could limit individual differences in this domain. Second, prepared food as a domain is more variable in shape, color and texture than the images for most of the other domains (e.g., cars, birds, faces) for which domain-specific ability have been measured. As such, food could be too heterogenous perceptually to correspond to a specific ability. For instance, it can be very difficult to reliably separate food from non-food images on the basis of simple low-level features such as spatial frequency or color (Tsourides et al., 2016). This research may therefore extend prior work on the measurement of visual abilities by demonstrating how methods developed in simpler domains may apply to the category of prepared food.

Based on our work linking domain-specific ability to experience in other domains (Bukach et al., 2010; Chang & Gauthier, 2021; Sunday et al., 2018), and on the link between FN and odor identification (Knaapila, 2017), we expect a negative correlation between FN and food recognition ability. We had no strong expectations about the association between food recognition ability and other aspects of with personality and attitudes towards food. There is some evidence that different aspects of personality are related to eating choices. For instance, more extraverted people seem to make eating decisions that are less healthy, while more conscientious people make healthier food choices (Keller & Siegrist, 2015). Openness may be related to exposure to a larger variety of foods, which could facilitate their recognition. Among the 5 classic personality dimensions, openness tends to be most strongly associated with FN (correlation of -.23 when openness does not include questions about food, (Knaapila et al., 2011; see also Nezlek & Forestell, 2019).

In summary, we create new measures of visual recognition of images of prepared food, seek to characterize performance on these tasks in terms of domain-general and domain-specific components and ask whether a visual ability specific to prepared food is related to FN. This work would greatly extend our understanding of the visual correlates of FN, which is more generally related to gustatory and olfactory functions, opening the door to studying whether poor visual abilities could play a role in sustaining FN.

2. Methods

2.1. Transparency and openness statement

We report how we determined our sample size, all data exclusions (if any), all inclusion/exclusion criteria, whether inclusion/exclusion criteria were established prior to data analysis, all manipulations, and all measures in the study.

2.2. Participants

We chose our sample size to ensure sufficient power to address hypotheses involving the correlation of o with food recognition, as well as testing the partial correlation between the two food tasks once controlling for o . Across different domains like musical notation (Chang & Gauthier, 2021) and chest nodule detection (Sunday et al., 2018), the correlation between o and different visual tasks was around $r=.35$. A sample size of 46 is required to detect such a correlation with a one-tail test, alpha of .05 and power of .80. We collected data from approximately three times this many participants to improve precision in our estimation of effect sizes..

We recruited participants fluent in English, located in the US, between the ages of 18 and 55 and who had never participated in studies from our laboratory on the Prolific.co

platform. We collected data from 135 participants. We excluded anyone self-reporting with a diagnosed or suspected eating disorder – 2 participants were excluded for this reason. Of the remaining 133 participants, we examined performance on the four visual tasks for evidence of not understanding the instructions. Three participants were at chance on one task but had very strong performance on the other 3 tasks, a pattern that is atypical and suggests a problem with a specific task. This was not a predetermined criterion but is relatively conservative as it leads to the exclusion of only 2% of participants. With a sample size of 130, our achieved power at an alpha of .05 was .78, .90 and .97 respectively to detect effect sizes of .20, .25 and .30.

We report data collected in February of 2022 for 130 participants, 67 female, 63 male, with a mean age of 33.7 (SD=9.12). Their self-reported ethnicity was Asian (11), Black (8), Hispanic (7), Multiracial (4), Native (1), Other (1) and White (98).

Ethical approval for the involvement of human subjects in this study was granted by Vanderbilt University Institutional review board, Reference number 050082. Participants gave informed consent via a form that included the statement “By clicking on the “Accept” button, you indicated that you agree to participate in the study and that you understand the information in this consent form. You agree that you are at least 18 years of age. You have not waived any legal rights you otherwise would have as a participant in a research study.” where an affirmative reply was required to enter the study. They were informed that the only identifying information kept based on their participation was their Prolific ID. They were financially compensated for their participation in the amount of \$3.75 (an average of \$9.13/hr).

2.3. Procedure

Participants were tested online after we verified that they were not participating on a mobile device. All participants were tested in a fixed order of tasks, including the order of trials within each task. They first answered a survey that included questions about general interest in visual recognition, interest about food, a short Big Five personality inventory (Donnellan et al., 2006) and the Food Neophobia Scale (FNS, (Pliner & Hobden, 1992). The surveys and tasks are described below, in the order of their completion.

2.4. Measures

2.4.1. General interest in visual recognition and Food interest survey

Three domain-general questions were adopted from Gauthier et al. (2014). Thirteen questions about interest and attitudes towards food followed. These questions about general interest in visual processing and about interest in food are presented in Table 1¹.

¹ One additional question was on the survey for some of the participants , and was removed due to experimenter error, so we removed it from all analyses. “Compared to other people, how varied is your diet on a day-to-day basis? (1: as little variety as possible; 7: as varied as possible).”

Table 1. General interest in visual recognition and Food interest survey	
item	Question and likert scale
GI-1	Generally speaking, how easily do you learn to recognize objects visually? (1: much less easily than average; 7: much more easily than average)
GI-2	Generally speaking, how strong is your interest in classifying objects in their various subcategories (e.g., learning about different kinds of insects, plants, shoes, tools, etc.)? (1: not strong at all; 7: very strong).
GI-3	Generally speaking, relative to the average person, how much of your time involves recognizing things visually? (1: much less time than average; 7: much more time than average).
FI-1	If you were asked to write an essay about different kinds of food, how extensive and detailed do you think your essay would be? (1: very short and simple; 7: very long and detailed)
FI-2	If you saw a specific food dish in a TV show, how sure are you that you could recognize that item among similar images if you were tested the next day? (1: not sure at all; 7: absolutely sure)
FI-3	How important is the domain of food to you, relative to all the other things you are interested in? (1: one of the least important; 7: one of the most important)
FI-4	How often do you read text (books, magazines, or online) that contain information about food? (1: almost never; 3:~once a month; 5:~once a day; 7: almost every day)
FI-5	How often do you try new foods? (1: almost never; 7: as often as possible)
FI-6	How likely are you to choose to wait longer or pay a little more for your food to taste better? (1: not likely at all; 7: very likely)
FI-7	How often do you talk about the taste of food when you talk to someone else? (1: almost never, 7: extremely often)
FI-8	If you wrote a review about a restaurant visit, how likely would you be to mention the taste of food, as opposed to other aspects of your visit? (1: not very likely; 7: extremely likely)
FI-9	How important is the visual appearance of food to you? (1: not important; 7: very important)
FI-10	How likely are you to choose to wait longer or pay a little more for your food to look better? (1: not very likely; 7: extremely likely)
FI-11	How often do you talk about the appearance of food when you talk to someone else? (1: almost never; 2: extremely often)
FI-12	If you wrote a review about a restaurant visit, how likely would you be to mention the appearance of food, as opposed to other aspects of your visit? (1: not very likely; 7: extremely likely)
FI-13	How often do you take pictures of food to share with others? (1: almost never; 7: extremely often)

2.4.2. Mini IPIP Scale

We used the Mini IPIP scale developed by Donnellan et al. (2006) to estimate personality on the Big Five dimensions. It is a 20-item short form that includes four statements to measure the five traits: intellect/imagination (openness), conscientiousness, extraversion, agreeableness, and neuroticism. Participants indicated how well each statement (e.g., “I have a

vivid imagination.” or “I get chores done right away”) described them on a scale from 1 to 5 (1: not very well; 5: very well).

2.4.3. Food Neophobia Scale (FNS)

We used the FNS (Pliner & Hobden, 1992) to measure food neophobia. The scale has 10 items, and participants indicated how well each statement (e.g., “I don’t trust new foods”, “I am afraid to eat things I have never had before”, “I will eat almost anything (R)”) on a scale from 1: disagree strongly to 7: agree strongly.

2.4.4. Food recognition ability

To measure food recognition, we chose to create two different tasks that would both tap into the ability required to match different images of the same dish but each would have different task demands. We therefore use the principle of aggregation (Rushton et al., 1983) which states that the average of multiple measurements will be more stable and representative than any single measure. For both our food recognition tests, we targeted the category of “prepared food”, loosely defined as items that go beyond single ingredients and that may be ordered from a menu or served at home as a single course (as part of a larger meal for breakfast, lunch or dinner). They may be served on a plate, in a bowl or even in a glass. The various dishes and images were selected to tap into the range of visual experiences that individuals in our US population may have with food at home, in restaurants, on television and online. The individual trials were selected to provide a broad range in terms of their difficulty (e.g., by selecting foils that varied in their perceptual similarity with the target items). An example of a relatively easy distinction could be a dish with noodles vs. rice, while a more difficult distinction would be between two different kinds of noodles, flat vs. round). The food

categories used on the tests (and listed in supplemental materials) included typical American dishes (e.g., mac and cheese or carrot cake) as well as dishes from ethnic cuisines that would be more or less familiar to American participants (e.g., tandoori chicken or plantain mofongo).

2.4.4.1. Food matching

This was a new test that included 1 example trial in which the answer was provided, followed by 60 trials. On each trial, an image of prepared food appeared for 1000 ms, followed by a set of three images of prepared food (Figure 1). Participants had to choose the image that was “most similar to the studied image”, without time restriction. There were no other instruction as to what was meant by similar (although the normative correct answer could be summarized as the dish that would receive the most similar label). Individuals could possibly use different understanding of what similar meant, but the first 10 trials included “correct/incorrect” feedback, providing some guidance. Trials were created to vary in difficulty. At the easiest level, the correct choice was created by cropping the same image of the target food in a different way. At the most difficult level, the correct choice would be an entirely different instance of the same food, presented in a different dish, from a different perspective. Foils were also selected to capture some visual features of both the target food and the table or plate. One additional trial was included as an attention check (it said “click here” in one of the panels).

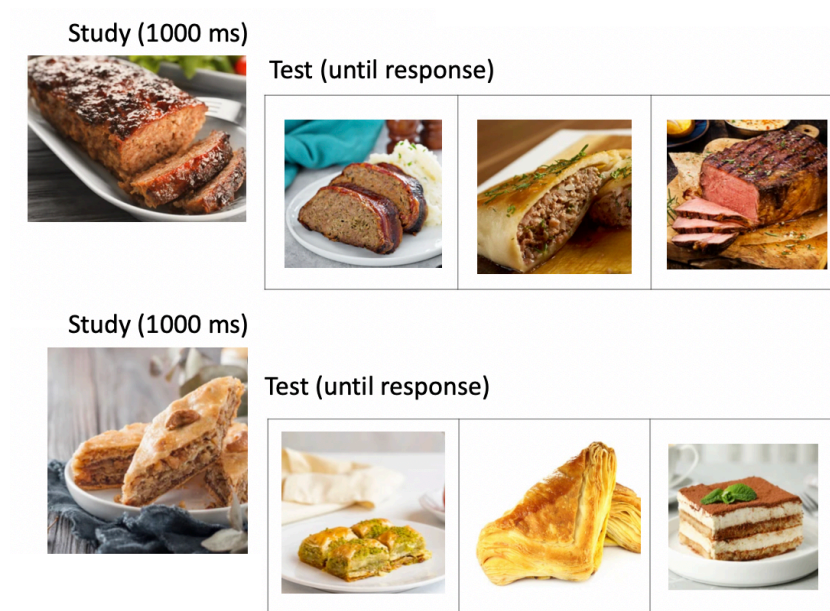


Figure 1. Two example trials similar to those in the Food matching test, using royalty-free images (correct choices are the first options in both cases.)

2.4.4.2. Food oddball test

This was a new test. Images of food were collected in the same way as for the food matching task (different dishes). On each trial, four images of food were presented, three of them showing different versions of the same dish, and one a different dish that varied in its visual similarity to the others, depending on the trial difficulty. Participants selected the oddball on each trial, without any time restriction (Figure 2). There was 1 practice trial and 60 trials on the test, with the first 5 providing correct/incorrect feedback. One additional trial was included as an attention check (3 of the images showed shoes).



Figure 2. Two example trials similar to those on the Oddball Food Matching test, using royalty-free images. Correct answers are the fourth and second, respectively.

2.4.5. Visual tasks to measure σ

Again, applying the principle of aggregation (Rushton et al., 1983), we used an approach that was successful in other studies (Chang & Gauthier, 2021; Chow et al., 2021; Sunday et al., 2018) and selected a memory task and a perceptual task, both using novel objects, to better capture a domain-general object recognition that is less specific to the demands of one task. Novel objects provide a good way to tap into domain-general abilities because we know that no variance on these tasks can come from experience with these objects.

2.4.5.1. NOMT-symmetrical Greebles

We used the Novel Objects Memory Test with one specific category of novel objects that share a common arrangement of parts (symmetrical Greebles, Richler et al., 2017).

Greebles are non-sense artificial objects used for visual studies (see Figure 3; Gauthier & Tarr,

1997). There were 48 trials of this visual task (Figure 3A). Participants initially studied six different Greebles presented together on the screen for 20 sec. This was followed by 6 3-alternative forced choice trials with 3 Greebles shown on the screen until one was selected as 1 of the 6 studied targets. The 6 study Greebles were then shown for study for another 20 sec, followed by 18 3-AFC trials showing objects in the same viewpoint as the targets, with the last 6 of these trials including gaussian noise superimposed on the objects. After another 20 sec of study with the 6 target objects, another 24 3-AFC trials followed, with objects in a slightly different viewpoint, and the last 12 of these trials including noise. One additional 3-AFC trial was included as an attention check, with foils from different object categories and in a different color from the Greebles, which were green.

2.4.5.2. Novel object matching test

The Novel Objects Match Test (adapted from Sunday et al., 2018) included 4 practice trials (with correct/incorrect feedback) and 70 trials on the test. On each trial, following a fixation cross shown for 500 ms, one image of a novel object was shown for 300 ms, followed by a category-specific mask for 300 milliseconds, and then a second image appeared with buttons for “same” and “different”. Participants had to judge whether the second object was the same as the first, independent of size or viewpoint (Figure 2B). There was a 4000 milliseconds to respond and the answer scored as incorrect if no response was provided. There was no feedback provided. Novel objects were from 4 different categories used in prior work (for reference, they are called asymmetrical Greebles, vertical Ziggerins, Horizontal Ziggerins and Sheinbugs – examples of two of these categories are shown in Figure 2), with each category in a distinct color and all trials showing either the same object or a distractor from the same

category. There were 31 same trials and 39 different trials. One additional trial was included as an attention check (it said “click here” in one of the buttons).

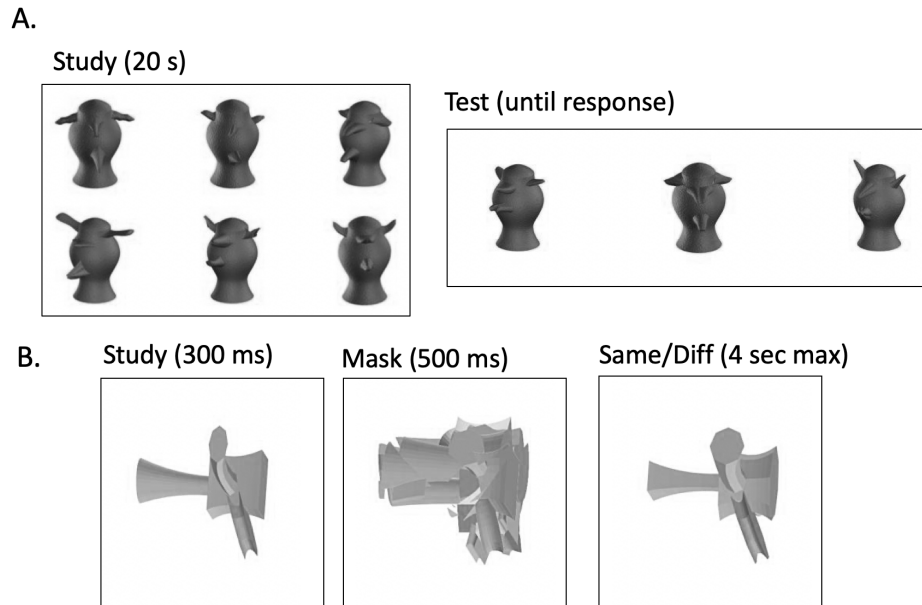


Figure 3. A. Study array and one sample trial for the Novel Object Memory test with symmetrical Greebles. Correct answer is the middle option. B. Sample trial for the Novel Object Matching test, using horizontal Ziggerins. Correct answer is different. In the study, the novel objects were shown in a single color for each category, such that color was not diagnostic.

2.5. Statistical analysis

Most statistical analyses were conducted in JASP 0.16.3 (JASP Team, 2020). Reliability was calculated with McDonald’s omega based on a single factor model (McDonald, 1999). Omega does not assume that every test item equally contributes to the factor measured by the test, unlike the more common Cronbach’s alpha. All surveys were scored using the mean of responses, except for FNS which is scored as a sum by convention. All visual tasks were scored as percent correct, except for the Novel Object Matching Test which was scored with sensitivity (d'), computed using $Z_{\text{hit rate}} - Z_{\text{false alarm rate}}$, adjusting for hit rates of 1 or false alarm rates

of 0 by adding .5 to the number of hits or false alarms and adding 1 to the number of same and to the number of different trials. In cases where we wanted to test support for a null result, we provided Bayesian tests using the default uninformative prior in JASP to calculate the BF_{01} values. This test shows how many times more likely the data are under the hypothesis of no correlation, relative to that of a correlation.

3. Results

3.1. Survey, mini-IPIP, FNS

Table 2 presents descriptive statistics for the survey measures, including reliability (internal consistency), which was generally high with least reliable test being the General Interest in visual recognition scale. This was expected as it had only had 3 items. It is notable that the mean FNS (28.5) is lower than that obtained in some recent US studies (e.g., the means were 30.5, 33.8 and 34.5 in 3 large surveys discussed in Jaeger, Chheang, et al., 2021). Differences in our sample include a narrower age range (18-55 rather than 18-65) and the fact that we excluded individuals with eating disorders. Table 3 provides the zero-order correlations between these measurements. We note that Bonferroni correction for 45 comparisons would yield a corrected alpha level of .001. Our new scale of interest in Food was correlated positively with general interest in visual recognition, agreeableness, extroversion and imagination, and negatively with FNS. In this sample, agreeableness and extroversion were most strongly (negatively) correlated with FNS, and imagination showed a more modest correlation in the same direction.

Table 2. Descriptive statistics for survey scores

	Mean	SD	min	max	McDonalds's ω	Skewness	Kurtosis
General Interest	4.87	0.88	2.67	7	0.68	0.26	0.1
Food Interest	4.33	1.1	2	6.8	0.88	0.15	-0.5
FNS	28.51	12.1	10	65	0.91	0.47	-0.39
Agreeableness	3.96	0.9	1	5	0.85	-1.08	1.01
Conscientiousness	3.76	0.79	1	5	0.77	-0.67	0.39
Extroversion	2.65	1.1	1	5	0.9	0.24	-0.9
Imagination	3.91	0.9	1	5	0.8	-0.82	0.25
Neuroticism	2.61	0.92	1	5	0.78	0.58	0.11

Table 3. Zero-order Pearson correlations for demographic and survey results

	1.	2.	3.	4.	5.	6.	7.	8.	9.
1. Gender									
2. Age	.01								
3. General Interest	.14	.10							
4. Food interest	.18*	.28**	.53***						
5. FNS	-.09	.02	-.08	-.35***					
6. Agreeableness	.24**	.06	.27	.44***	-.31***				
7. Conscientiousness	-.01	.12	.13	.17	-.12	.05			
8. Extroversion	.13	.11	.23**	.49***	-.32***	.46***	.03		
9. Imagination	.01	.14	.46***	.33***	-.21*	.40***	.09	.29**	
10. Neuroticism	.11	-.29	.02	-.12	.07	.03	-.44***	-.21*	-.11

Note: * $p < .05$, ** $p < .01$, *** $p < .001$.

3.2. Visual tasks

Table 4 presents the descriptive statistics for the visual tasks, including for the two aggregate indices for o and Food Recognition Ability. We estimated o using the aggregate of the z-scores for the NOMT and Object Matching tasks ($r = .31$, $p < .001$), and food recognition ability using the aggregate of the z-scores for the Food Matching and Food Oddball tasks ($r = .61$, $p < .001$). The correlation between o and food recognition ability was $r = .40$, $p < .001$ (Figure 4), providing evidence that some of the variance captured in our food recognition tasks can be attributed to a domain-general ability. The partial correlation between the two food tasks, controlling for o , was $r_{\text{partial}} = .55$, $p < .001$ (Figure 5). This illustrates that not all of the shared

variance in these two food recognition tasks can be accounted for by σ , providing evidence that we are measure a food-specific recognition ability.

Table 4. Descriptive statistics for visual tasks and aggregate scores

	Mean	SD	min	max	McDonalds's ω	Skewness	Kurtosis
NOMT-SymGreeb (acc)	0.53	0.13	0.27	0.90	.72	0.42	0.20
Object matching (d')	0.98	0.65	-0.49	2.87	.76	0.18	-0.41
Food matching	0.79	0.11	0.40	0.98	.82	-1.12	1.40
Food oddball (acc)	0.62	0.13	0.18	0.93	.85	-0.41	0.33
Food ability (z-score)	0	0.81	-2.69	1.88	.90	-0.67	0.51
O (z-score)	0	0.90	-1.68	2.13	.80	0.20	-0.47

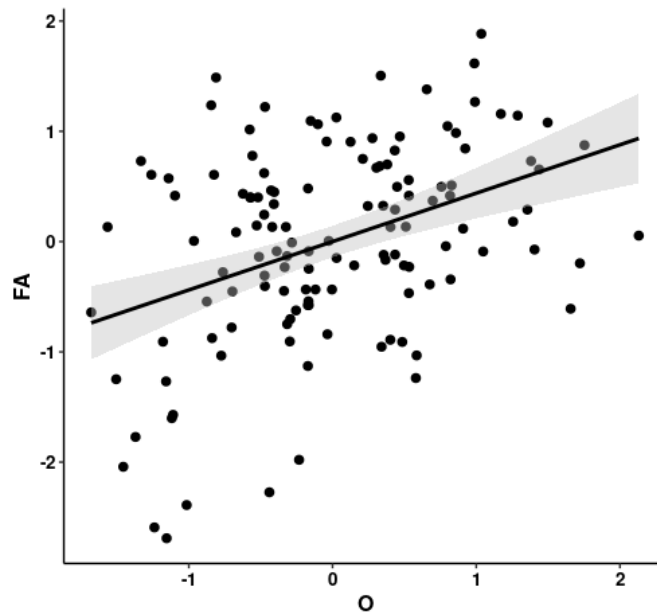


Figure 4. Correlation between σ and food recognition ability (both variables are z-scores). Shaded area shows 95% confidence interval.

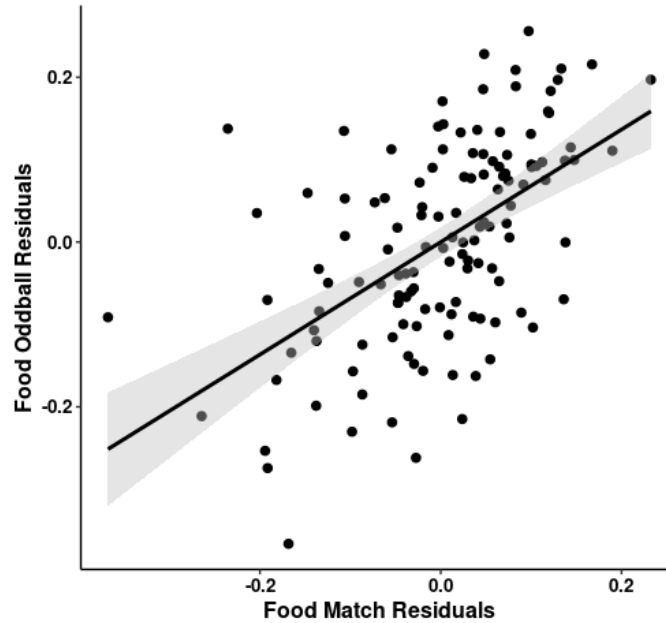


Figure 5. Partial correlation between the Food Matching and the Food Oddball tests, controlling for o (both variables are z-scores). Shaded area shows 95% confidence interval.

3.3. Other predictors of food recognition ability, including food neophobia.

We explored which aspects of attitudes about food or personality may relate to food recognition ability after controlling for o (that is, we aim to see what predict the domain-specific variance that contributes to our visual food tasks). A Bonferroni correction for 10 comparisons would yield a corrected alpha level of .005. At this corrected level, only FNS correlated negatively with food recognition ability residuals, $r_{\text{semipartial}} = -.31$, $p < .001$ (Table 5 and Figure 6).

Table 5. Partial correlation between Food recognition ability and demographic and survey measures, controlling for O.

	Food ability residuals
Age	-.05
Gender	.20*
General interest	-.18*
Food interest	.02
FNS	-.31***
Agreeableness	.02
Conscientiousness	-.03
Extroversion	.00
Imagination	-.05
Neuroticism	.14

Note" * $p < .05$, *** $p < .001$

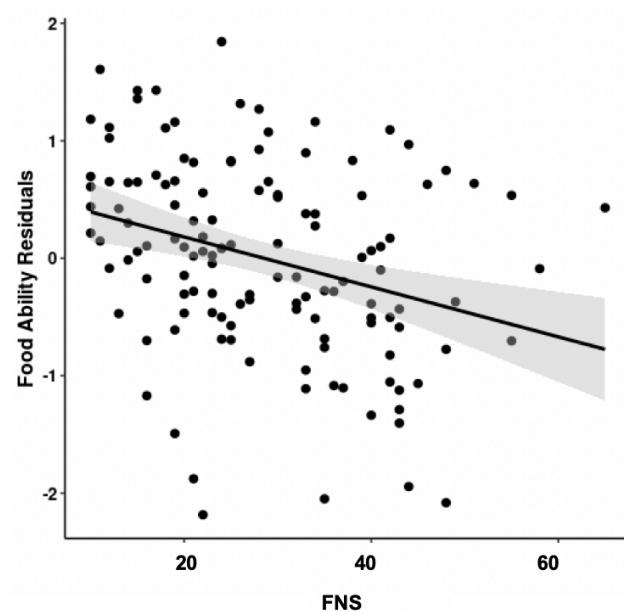


Figure 6. Semipartial correlation between FNS scores and the Food Recognition Ability, controlling for o. Shaded area shows 95% confidence interval.

We also conducted a hierarchical regression on food recognition ability, entering *o* in the first step and the other predictors simultaneously in a second step (Table 6). This analysis was very similar to individual partial correlations, suggesting little influence of multicollinearity. The second step in the model accounted for 15% of the variance in food recognition ability. The largest contribution was the negative contribution from FNS scores. In addition, women showed an advantage in food recognition ability, which was also predicted by general interest in visual discriminations, even when *o* was controlled. This last result suggests that general interest in visual discriminations is not relevant to visual tasks with novel objects (indeed, the correlation with *o* was not positive, $r = -.15$, $p = .08$), but instead reflects interest in more realistic applied domains.

Table 6. Hierarchical regression predicting Food Recognition Ability (standardized coefficient values in the table).

Predictor	Step 1	Step 2
O	.40***	.40***
Age	-	.01
Gender	-	.21*
General Interest	-	-.27*
Food interest	-	.07
A	-	-.10
C	-	.03
E	-	-.04
I	-	.04
N	-	.17
FNS	-	-.33***
Adj.R-square	.15	.28
R-square change	-	.15

Note. * $p < .05$, *** $p < .001$

Finally, we note that our visual tasks are designed to be RT-irrelevant (Draheim et al., 2019), in that the participants can take as long as they need to respond (the only task with a response deadline was the object Matching task, with a 4s deadline, which was more than sufficient as the mean of participants' median RTs in this task was 1025 ms, SD = 250 ms). The use of RT-irrelevant tasks is meant to increase reliability and avoid the need for complex models necessary to combine accuracy and RTs into single measures (Vandierendonck, 2017). Nonetheless, we analyzed RTs to test for the possibility that participants high in food neophobia may have chosen to spend less time looking at the images of food, which could have impacted their accuracy. However, FNS scores were not related to the median RTs neither on the Food Matching task ($r = .04$, $p = .67$, $BF_{01} = 8.34$) or on the Food Oddball task ($r = .08$, $p = .40$, $BF_{01} = 6.42$). The values we obtained provide “substantial” evidence for the null (Jeffreys, 1961). This suggests that avoidance of food images during the visual tasks is unlikely to explain our results.

4. Discussion

This work had three objectives: 1) to develop the first tests designed to measure individual differences in the recognition of images of prepared food; 2) to measure how much of the performance on such tasks was accounted by *o* vs. a food-specific ability; and 3) to test whether individual differences in food-specific recognition ability are related to FN.

To address the first goal, we created two tests to measure the visual ability to recognize prepared food. The two tests showed good reliability and our results lend validity to the domain-specific construct that we targeted. In particular, and relevant to our second goal, performance on the two food tasks was correlated even after correcting for *o* (the correlation

disattenuated for measurement error was .66). Importantly, o by itself also accounted for a substantial part of the variability in food recognition (the correlation between o and of food recognition, disattenuated for measurement error was .47). These results indicate that visual decisions about prepared food are similar to decisions about single objects (e.g., cars or birds) or even to medical decisions like those about detecting tumors in chest x-rays. People who excel in these tasks are generally likely to be good at recognizing food, but a considerable part of variability in visual performance with food is specific to this domain.

Do o and food-specific recognition ability interact? While we did not study this, we can speculate. Because o is measured with novel objects, it should be clear that it reflects an ability that facilitates visual decisions regardless of knowledge or familiarity. But a high o could also facilitate the acquisition of domain-specific knowledge that generalizes to unfamiliar items of the domain. In studies where individuals were trained to perceptual expertise with novel objects, this expertise transferred to new objects of the trained category (Gauthier & Tarr, 1997; Wong et al., 2009). Our tests did not measure the ability to name a dish, or indeed test for any aspect of knowledge about food and we did not measure familiarity with any of the individual dishes we presented. It is likely, although we did not verify this, that people with more experience with food are not only good at perceptual decisions about dishes they have experience with, but also in matching images of dishes they have little familiarity with. Future work measuring visual recognition and familiarity for specific dishes could answer this question.

Addressing our third goal, we found that food recognition ability, even after controlling for o , was negatively related to FN, as predicted. FN captures important attitude towards food that are likely to influence not only what one eats, but also one's exposure to visual images of

foods. We hypothesize that FN may influence food recognition ability through the mediating effect of exposure to foods, but it is equally possible that limited exposure to food reduces familiarity, which gains importance at increased levels of FN in driving food choices (Jaeger, Roigard, et al., 2021). The specific relationship of visual ability to variables such as novelty, familiarity and arousal (Jaeger, Chheang, et al., 2021) will need to be addressed in future research, ideally in longitudinal studies. Our food tasks will facilitate the measurement of the possible perceptual consequences of experience with food.

The association between FN and food recognition ability was specific and held when controlling for personality, gender and age. While our new measure of food interest was substantially negatively related to FN, it was not associated with visual food recognition and including it in the model did not reduce the effect for FN. It is possible that experience with food is so extensive even in individuals with relatively low levels of interest that only the kind of avoidance measured by FN affects visual learning in this domain. We note that across several domains, it has been surprisingly difficult to measure individual differences in experience over the course of a lifetime, in a manner that accounts for visual performance (Gauthier et al., 2014). The specific aspects of experience that are important to visual learning, be they expressed in quality or quantity, can vary a great deal depending on the domain. For instance, in one domain (e.g., musical notation) it may be useful to ask about years and daily amounts of practice but in a domain such as food, attitudes may be more relevant. For this reason, our understanding of how experience influences visual performance greatly benefits from experimental manipulations of experience (Chua & Gauthier, 2020; Gauthier & Tarr, 1997; Wong et al., 2009). While domain-specific visual training in food recognition could be

challenging because of the wide range of relevant items, training in the visual recognition within a sub-domain of the broader category (e.g., sandwiches or pasta dishes) could be more tractable. Whether such training would affect FN is an interesting question. Programs that use picture books to increase children's familiarity with unfamiliar fruits and vegetables have shown some success in increasing willingness to eat these foods (Heath et al., 2011). It is unclear how familiarization with individual ingredients would translate into perceptual abilities to match images of prepared food in adults. Our tests used more complex images of complete dishes rather than ingredients which possibly could be more effective in adults. Experimental studies comparing different kinds of training have shown that exposure is not sufficient to improve the kind of perceptual skill we measure here. Practice discriminating between visually-similar items appears necessary for this kind of perceptual expertise (Bukach et al., 2012; Wong et al., 2009). Likewise, it would be important to ascertain whether improving one's perceptual ability to recognize certain food visually could also generalize one's attitudes towards that food when described verbally (as on a menu). In any case, it is fortuitous that 30 years of work on the FNS (Rabadán & Bernabéu, 2021) and 30 years of progress in the measurement of visual abilities have allowed us to uncover this association between FN and food recognition ability.

We acknowledge several limitations of this research. First, the sample we used is a convenience sample and also limited in size. We found small effects of age and sex on food recognition ability, but our sample size was too small to further explore these effects. Our results are promising but should be replicated in larger representative samples of the general population. Second, our measurement of personality was based on a very short scale that may not have captured the important aspects of openness as relevant to FN. Third, our

measurement of “food interest” was based on a new survey that was not validated separately and that may not reflect a unidimensional construct (a larger sample would be needed to explore this question). Finally, we chose to exclude individuals with eating disorders, because they are relatively rare in the general population (Smink et al., 2012) and we were specifically interested in the role of FN as a stable trait. There are similarities and differences between picky eating, food neophobia and avoidant/restrictive food intake disorder (Mammel & Ornstein, 2017) and these considerations were beyond the scope of the present project. It is our hope that the measurement of food recognition ability may support further research in this area. This could include exploring the possibility that increasing the perceptual ability to recognize food, as in effective training protocols in other domains (Gauthier & Tarr, 1997; McGugin et al., 2011; Wong et al., 2009), could help interrupt the cycle through which FN impacts dietary choices.

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